Bindu Sree Kavuru

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PROFESSIONAL SUMMARY

Results-driven Data Scientist with **3+ years of experience** building predictive models, designing scalable data pipelines, and delivering actionable insights across enterprise and mission-driven environments. Skilled in **Python, SQL, ML algorithms, TensorFlow, Scikit-learn, data preprocessing, and cloud-based analytics**, with a proven ability to optimize model performance and automate data workflows. Experienced in transforming complex datasets into high-impact solutions that support decision-making and operational efficiency. Adept at collaborating with cross-functional teams to deploy reliable, ethical, and scalable machine learning systems.

EDUCATION

University at Buffalo, *Master of Science in Data Science* | NY, USA **August 2023 – May 2025**

MRECW, *Bachelors in Computer Science* | TS, India **June 2017 – June 2021**

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Cisco Certified Network Associate (CCNA)
- Oracle Cloud Infrastructure 2025 Data Science Professional
- Oracle Certified Associate (Java SE8)

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, C, C++, SAS
- **Visualization Tools:** Power BI, Tableau, Plotly, Streamlit, EDA, Palantir
- **Project Management:** JIRA, Scrum, Kanban
- **Cloud Platforms:**
  - **AWS:** S3, Lambda, EMR, SageMaker, Redshift, Athena, EC2
  - **Azure:** Data Factory, Data Lake, Databricks
  - **GCP:** BigQuery, GCS, Cloud Functions
- **DevOps & CI/CD:** Docker, Kubernetes, GitHub Actions, CI/CD Pipelines, Airflow
- **Big Data & ETL:** Hadoop, Spark, Kafka, Hive, Delta Lake, Data Analysis & Manipulation
- **Data and ML:** Scikit-learn, XGBoost, Tensorflow, Jupyter
- **Machine Learning and AI:** Gen AI, Large Language Models
- **Databases & Warehouses:** PostgreSQL, MySQL, MongoDB, DynamoDB, Oracle, SQLite, Microsoft SQL Server

WORK EXPERIENCE

Data Scientist | Apiezy **August 2025 - Present**

Technical Skills: Python, SQL, OpenAPI/Swagger, JSON, Pandas, Data Profiling, Feature Engineering

- Analyzed OpenAPI and Swagger specifications to extract schemas, request/response structures, and metadata essential for building ML-ready datasets supporting predictive modeling and LLM workflows.
- Engineered Python- and SQL-based pipelines to clean, standardize, and transform API-related data, improving dataset consistency by 45% and enabling reliable downstream analytics.
- Built structured feature datasets derived from API endpoints, logs, and parameter patterns to support ML model training, validation, and automated quality checks.
- Designed data-profiling routines to evaluate schema coverage, detect anomalies, and identify feature drift, strengthening the accuracy and stability of machine learning pipelines.
- Collaborated with engineering and QA teams to clarify data flows, resolve inconsistencies in API behavior, and enhance the usability of extracted metadata for analytical and modeling tasks.
- Documented data transformation logic, schema mappings, and feature-engineering processes to improve ML workflow reproducibility and accelerate onboarding for cross-functional teams.

Graduate Assistant | University at Buffalo **August 2024 - May 2025**

Technical Skills: Python, SQL, Pandas, NumPy, Scikit-learn, Tableau, AWS Quicksight

- Analyzed complex datasets using Python and SQL, applying machine learning and statistical modeling techniques to improve fraud-detection workflows, achieving 85% model accuracy across multiple evaluation cycles.
- Designed Tableau and Quicksight dashboards to visualize model outputs and trend analyses, enabling stakeholders to interpret results quickly and reducing decision-making time by 30% through clear, data-driven insights.
- Built scalable data pipelines using Pandas and NumPy, integrating ETL steps and automated validation checks to enhance data quality and reliability for research-driven analytics projects.
- Developed feature-engineering scripts and preprocessing routines that improved model readiness, reducing manual preparation efforts and strengthening reproducibility for ongoing ML experiments.
- Collaborated with faculty and research teams to document analytical workflows, streamline experimentation processes, and support deployment of AI-driven solutions in academic settings.

Data Scientist II | Accenture Pvt. Ltd. **November 2022 – July 2023**

Technical Skills: AWS Lambda, DynamoDB, Apache Kafka, Kubernetes, Python, SQL, PyTest, CI/CD

- Engineered fraud detection models using Python, SQL, XGBoost, and Scikit-learn, reducing false positives by 20% and losses by 12% across 1M+ monthly transactions, with real-time AWS Lambda alerts for AI integration.
- Optimized data pipelines with Apache Kafka, Spark, and Docker, applying statistical methods to process 500,000+ records and identify trends, saving 500+ analyst hours annually through automated ETL processes and collaboration with data engineering teams.
- Analyzed and documented legacy PL/SQL code for a healthcare client, mapping interlinked procedures and functions across 20-year-old datasets to facilitate migration to Java, ensuring data integrity and compliance while tracking progress in JIRA.
- Developed interactive Tableau dashboards and visualizations to communicate insights from complex datasets to non-technical stakeholders, improving decision-making efficiency by 18% and supporting anomaly detection in intelligence workflows.
- Implemented machine learning solutions with TensorFlow and Python to address system anomalies, enhancing model robustness by 18% under tight deadlines, and validating performance through cross-functional team coordination.
- Contributed to scalable AI architectures using Kubernetes and CI/CD pipelines, integrating relational databases like DynamoDB for large-scale data handling and real-time analytics in fraud mitigation projects.

Associative Data Scientist | Accenture Pvt. Ltd. **July 2021 – November 2022**

Technical Skills: AWS Lambda, DynamoDB, Apache Kafka, Kubernetes, Python, SQL, PyTest, CI/CD

- Analyzed 500,000+ records using SQL and Excel, extracting actionable insights to support fraud mitigation, reducing losses by 12% over 14 months through statistical analysis and stakeholder collaboration in intelligence workflows.
- Created Tableau dashboards to communicate complex results to non-technical teams, improving decision-making and operational efficiency, while supporting data engineering with Docker and relational databases for large-scale data handling.
- Implemented data-driven solutions with Python, addressing anomalies to strengthen systems, enhancing robustness by 18% and coordinating with teams to validate AI model performance under tight deadlines.

PROJECTS UNDERTAKEN

Chronic Disease Risk Prediction

Tech: Azure Data Factory, Spark, XGBoost, Python, Power BI

- Engineered an end-to-end ML pipeline for a 30,000-row synthetic patient dataset, achieving 92.7% accuracy in multiclass classification (7 disease classes: Diabetes, Hypertension, Obesity, Asthma, Arthritis, Cancer, Healthy) using XGBoost.
- Optimized data preprocessing with Azure Data Factory for ETL and Spark for scalable transformations, implementing derived columns and window functions to handle missing data.
- Developed a Power BI dashboard with dynamic patient-ID drilldown and per-patient accuracy cards (0/1 scoring), using slicers to sync visuals for stakeholder-driven insights, enhancing chronic disease risk management.

Buffalo Crime Data Dashboard with Live Updates

Tech: Streamlit, PostgreSQL, Python, Tableau, Pandas

- Developed a real-time crime analytics dashboard using Streamlit and Tableau, processing 100,000+ daily records from Buffalo's public crime dataset to deliver predictive insights for law enforcement and civic planning.
- Engineered an automated data pipeline with Python and Pandas, integrating PostgreSQL for scalable SQL queries and scheduling updates to reduce decision latency by 40% through optimized data retrieval.
- Designed interactive Tableau visualizations, including maps, heatmaps, and year-wise breakdowns, enabling stakeholder-driven analysis and improving resource allocation by 25% over 6 months.

AI-Powered SQL Query Assistant

Tech: Streamlit, PostgreSQL, Python, SQL

- Built an AI-driven assistant using Python and Streamlit, enabling users to upload datasets and query data via natural language prompts generating SQL queries and result tables with 98% query accuracy.
- Automated SQL query execution on PostgreSQL for user-uploaded datasets, processing 10,000+ queries monthly and reducing manual query time by 50% through AI-powered automation.
- Developed dynamic Tableau visualizations based on user prompts, delivering interactive dashboards with real-time data insights, enhancing stakeholder analysis for diverse datasets.